

MY457: Reappraisal of Polity size and local government performance: Evidence from India

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1. Research Design and Assumptions

The authors of “Polity Size and Local Government Performance: Evidence from India” investigate how the size of local political units influences the delivery of public services, using the state of Uttar Pradesh as a case study. Their research design leverages a natural experiment created by a government rule which required villages with populations of 1,000 or more, according to the decadal census to be designated as their own gram panchayat (GP) which is the lowest tier of elected government. In contrast, villages just below the 1,000 person threshold are grouped with others, resulting in larger GPs. This creates a sharp discontinuity in GP size around an arbitrary cutoff. The authors exploit this using a regression discontinuity (RD) design. By comparing villages just above and below the this threshold, they estimate the causal impact of being in a smaller versus a larger GP on a variety of public service outcomes which have been replicated successfully in this paper.

The data used in this study are drawn from multiple sources, including three rounds of the Indian Census (1991, 2001, and 2011), administrative records on public programs such as the National Rural Employment Guarantee Scheme (NREGS), village-level socioeconomic surveys (like SECC and Mission Antyodaya), local government budget data, and electoral records. This allows the authors to examine short and long-run outcomes related to infrastructure, social welfare programs, and political behavior across more than 100,000 villages over three decades.

The validity of this causal identification strategy relies on two main assumptions. First, there must be no manipulation of village populations around the 1,000-person cutoff. The authors provide evidence that this is unlikely due to the reliability of Indian census procedures and the fact that future delimitation was not anticipated during census enumeration. Second, all other characteristics of villages must be continuous around the cutoff. The authors show that observable covariates such as literacy rates, infrastructure presence, and caste composition (replicated here), do not change discontinuously at the threshold. Additionally, they conduct multiple robustness checks to rule out the possibility that other programs use the same population threshold for resource allocation.

Through this RD approach, the study finds that smaller GPs tend to deliver better public services, including improved education infrastructure, sanitation, welfare benefits, and workfare access. These gains appear to arise from increased civic engagement and more responsive local leadership, rather than mechanisms like elite capture or administrative efficiency. They claim their findings provide compelling evidence that reducing the size of local political units can improve governance outcomes in low-capacity settings, with broader implications for decentralization policy design.

Despite these strengths, one critical issue is the bundled nature of the treatment at the RD cutoff. While the authors claim that crossing the threshold provides variation specifically in polity size, in reality, this cutoff may affect multiple dimensions simultaneously such as geographic compactness, demographic composition, social homogeneity, and political dynamics within the GP. This bundling complicates interpretation because the estimated causal effects reflect multiple intertwined mechanisms rather than the sole impact of GP population size. Although the authors acknowledge this limitation briefly, they provide limited empirical

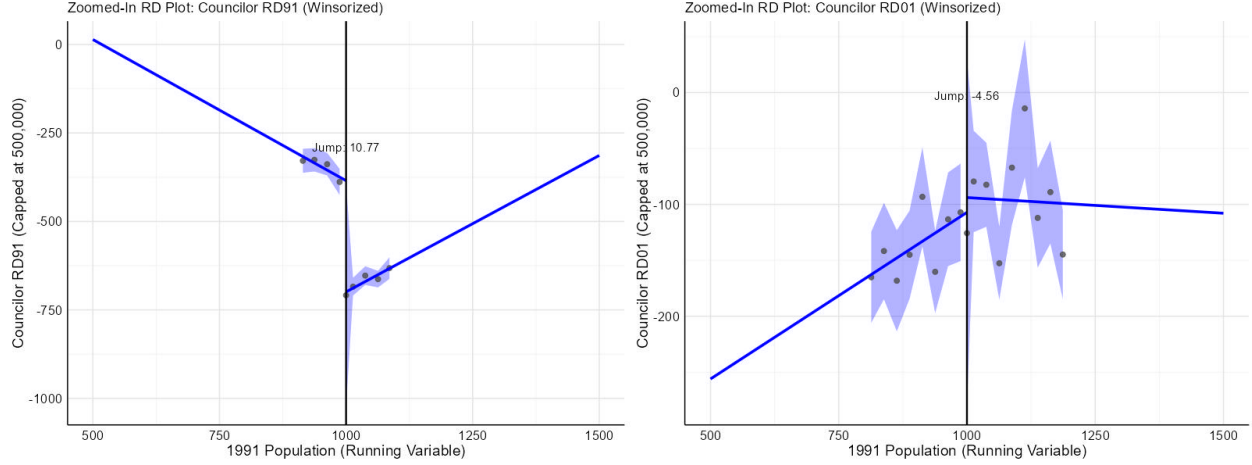


Figure 1: Political Representation Discontinuities at the 1,000 Population Threshold

investigation to disentangle these dimensions, potentially weakening confidence in the claim that the observed improvements are primarily attributable to smaller polity size alone.

The regression discontinuity plots provide some evidence for isolating specific mechanisms within this bundled treatment. Figure 1 shows sharp discontinuities in the number of ward councilors per capita at the treatment threshold, consistent with institutional rules linking population size to representation. This supports the hypothesis that enhanced political access rather than other bundled factors may be a key mechanism driving improvements in public service delivery. However, the data reveal some extreme outliers, including negative values as low as -2000, which likely reflect data entry errors or misprints.

Table 1: Summary Statistics by Caste Diversity

high_caste_diversity	n	mean_sc_share	mean_st_share	mean_total_caste_share	mean_population
0	25660	0.1157379	0.0000946	0.1158325	1042.8544
1	25657	0.3763755	0.0019021	0.3782776	1033.5466
NA	3	0.2895417	NaN	NaN	903.6667

Table 1, reveals minimal variation in caste diversity across different population thresholds, with SC population shares showing only slight differences and ST shares remaining relatively stable. This suggests that the governance improvements observed at smaller polity sizes cannot be attributed to more homogeneous caste composition.

Moreover, the balance tests confirm that the pre-treatment covariates such as Scheduled Caste (SC) and Scheduled Tribes (ST) population shares show only minor discontinuities around the threshold. This suggests that demographic composition changes are not the primary driver of observed effects. While these findings do not fully resolve the bundling problem, they help narrow the range of potential mechanisms and improve confidence that political representation, as opposed to demographic sorting or geographic factors, plays a central role in the observed governance improvements.

The significance of these balance tests becomes even more apparent when contrasted with findings from literature on social fragmentation and governance. Baland et al. [2008]’s work in India finds evidence that “heterogeneity in caste hinders collective action to manage firewood extraction and irrigation.” Given this established finding that even modest demographic heterogeneity can substantially undermine public goods provision, the absence of significant discontinuities in SC and ST population shares around the 1,000 threshold takes on heightened importance. The fact that governance improvements occur without meaningful changes in caste composition provides compelling evidence that political representation mechanisms, rather than demographic sorting, drive the observed benefits of smaller polity size. This demonstrates that decentralization

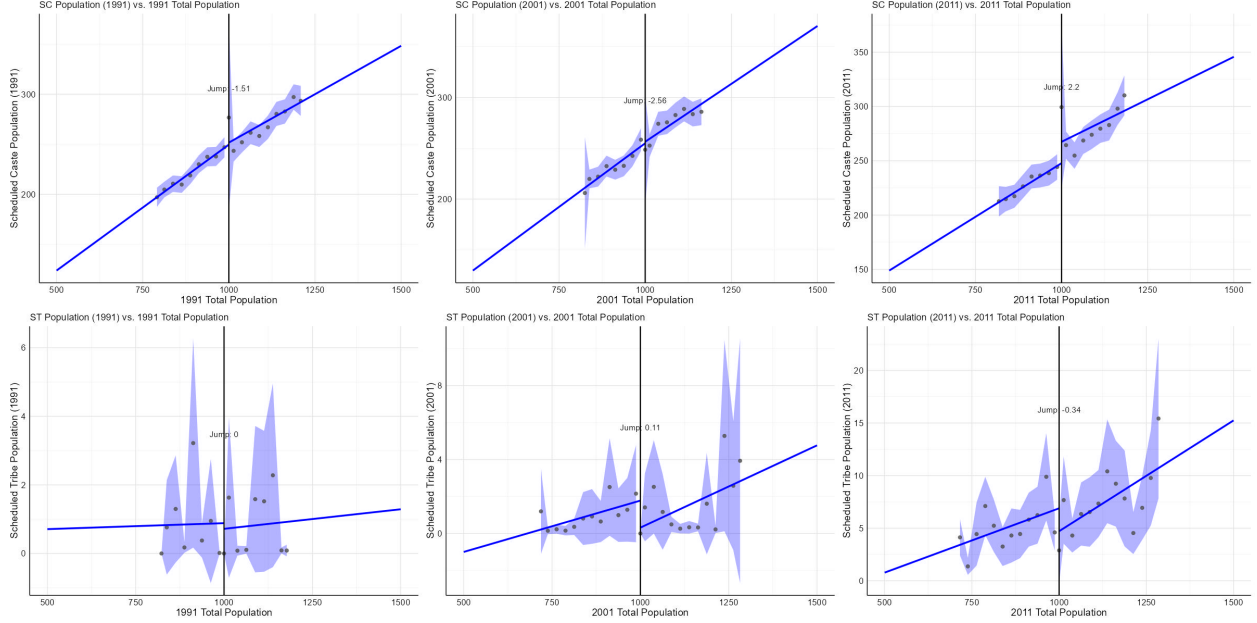


Figure 2: Balance Tests: Pre-Treatment Caste Composition Around the Population Threshold

reforms can improve governance outcomes even in socially heterogeneous contexts where fragmented communities are typically expected to struggle with effective public goods management, though other bundled mechanisms such as improved administrative efficiency or enhanced elite accountability may also contribute to these effects.

3. Replication

An attempt was made to replicate all figures from the original analysis. However, several figures relied specifically on the *rdpermute* and *rwolf2* Stata packages, neither of which have direct equivalents or available implementations in R. This created obstacles in replication and transparency. Researchers attempting to replicate findings using alternative statistical software (such as R) may face hurdles, potentially leading to inconsistencies or errors in reproducing results, however, that was not the case when Narasimhan and Weaver's paper.

In the replication package for `main_analysis`, specifically regarding Figure 1 (replicated as Figure 3 here), there is an error in the specification of the running variable. The original replication code references the state code variable `pc91_pca_tot_`, but this variable does not exist in the provided data set, `vilage_analysis_cleaned.dta` located in the `analysis_data` folder. Instead, the correct variable name should be `**pc91_pca_tot_p`. When substituting this correct variable (`pc91_pca_tot_p*`) into the R replication script, the generated results match those presented in the original paper as seen below. This confirms that the discrepancy was a typographical or coding oversight in the replication materials, rather than a substantive error affecting the analysis.

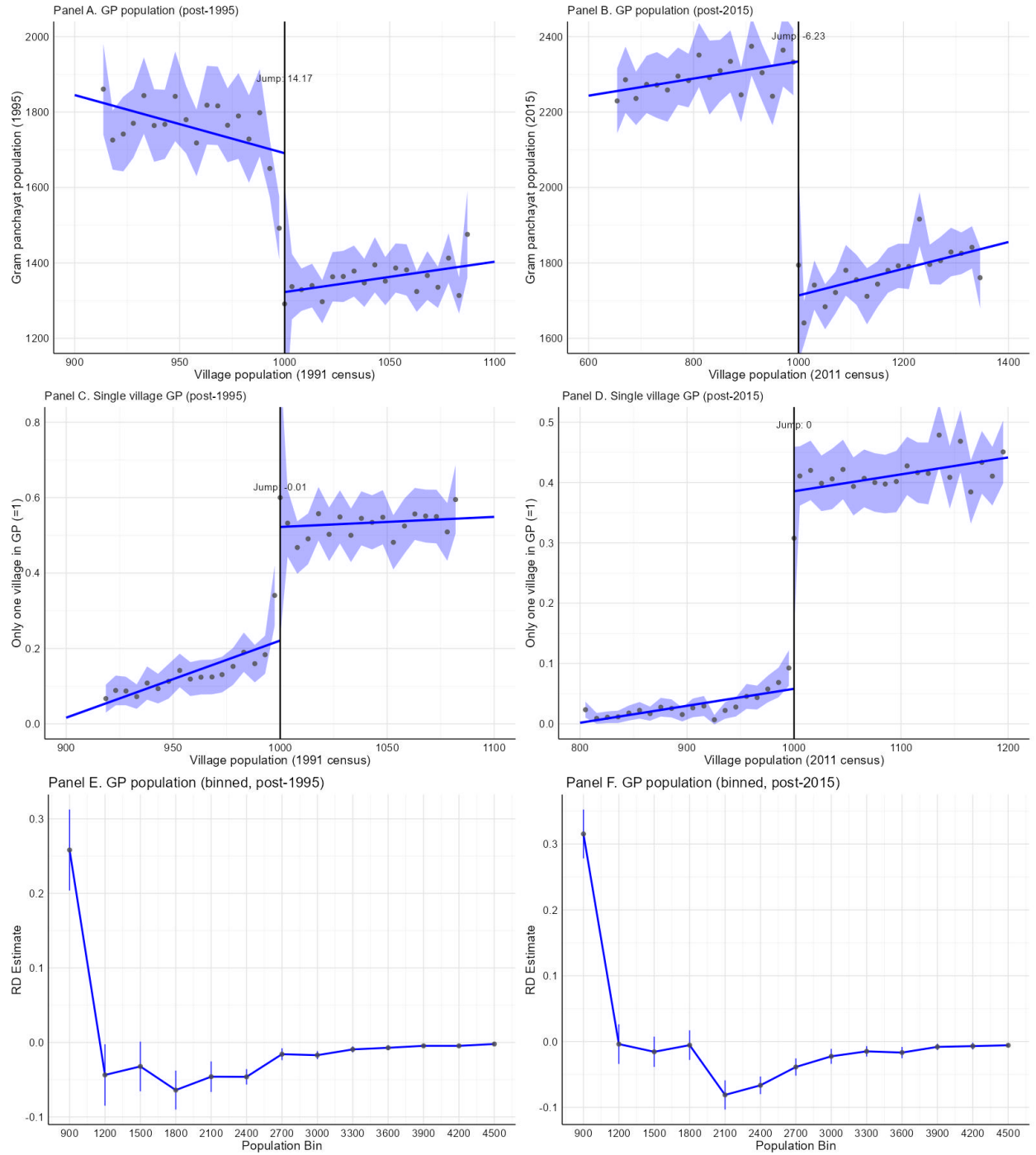
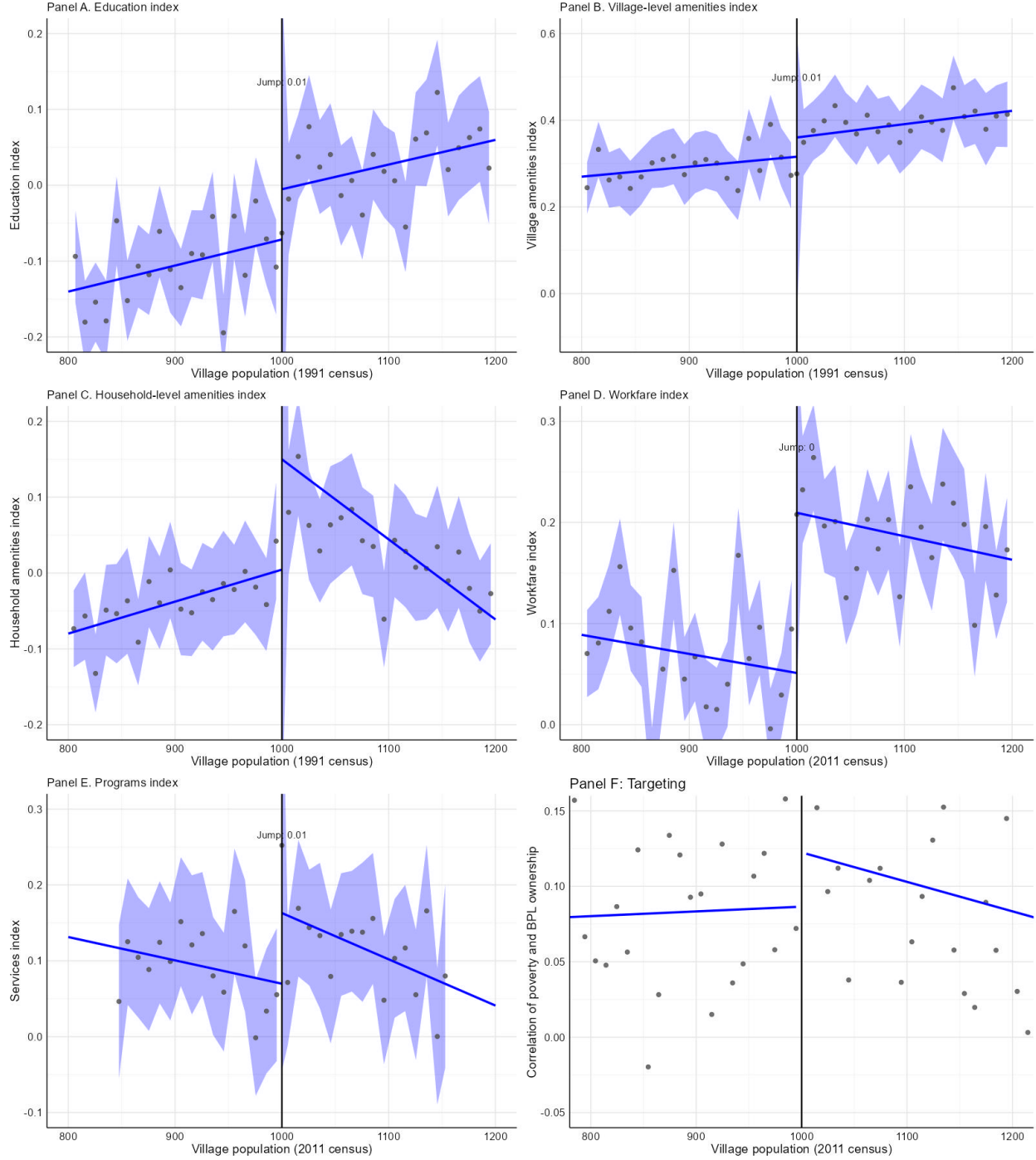
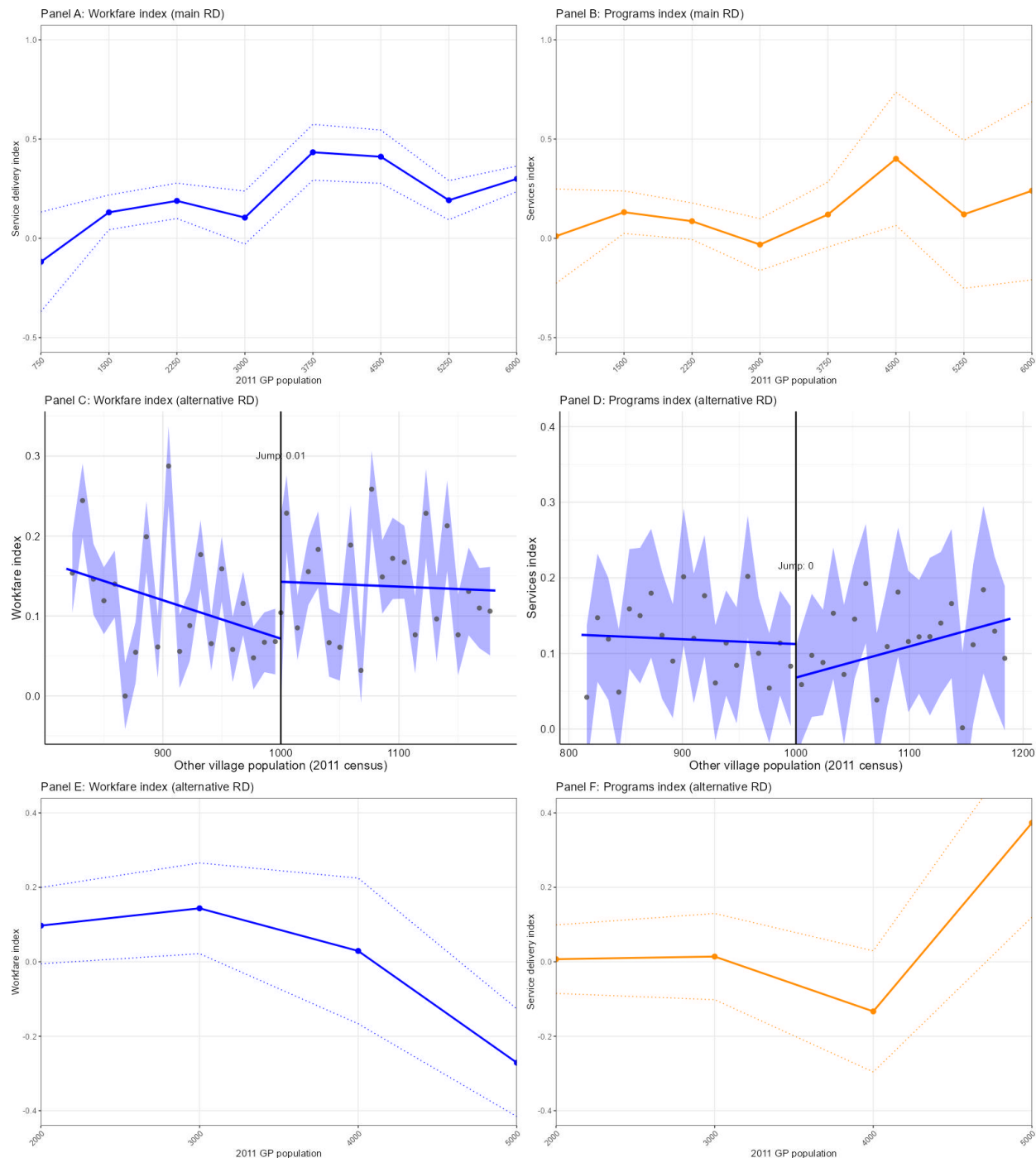


Figure 3: First Stage of Village Population on Gram Panchayat Population in Uttar Pradesh (Replicated)



As part of the replication process, the original Stata based regression discontinuity plotting function *rd-plot_custom* was re-implemented in R with several improvements aimed at enhancing flexibility, reliability, and interpretability. One of the key modifications involved the explicit renaming of parameters for greater clarity—for example, *c* was renamed to *cutoff* for instance. Unlike the Stata version, which implicitly determined axis limits, the R implementation introduces user defined *xlim* and *ylim* arguments,. This allows for precise control over the plotting window. This also allows researchers to standardize visual scales across multiple plots, thereby improving comparability. Moreover, the clustering functionality which is handled in Stata via *rdbootstrap* and *rdplot* was omitted in the R version due to limitations in R's *rdplot* implementation since it does not natively support clustering in the same way [Calonico et al., 2023]. Additionally, comparing

the R and Stata implementations reveals that the programs index and workfare index generated visibly smaller confidence intervals in the R plots than in the corresponding Stata outputs.



Another enhancement made to *rdplot_custom* relates to bandwidth selection and robustness. In the original Stata function, the user was allowed to choose a bandwidth selection method via the *bwselect* parameter. However, this choice was ultimately ignored, as the function hard-coded the use of *e(h_mserd)*, effectively defaulting to the MSERD method regardless of users input. The R version corrects this by calculating and averaging the left and right optimal bandwidths returned by *rdrobust* and introduces a fallback mechanism

where if either bandwidth is invalid by introducing a default value based on half the data range that is used. This prevented runtime errors that were common in the Stata version when bandwidth selection failed.

Replicating the *MultiPanelTabRD* function in R also required significant modifications and enhancements to ensure functional parity with the original Stata implementation. One of the key challenges arose from R’s stricter handling of NULL values and its reliance on vectorized operations, which led to errors when panel-specific parameters (e.g., *running_varB*, *cutoffB*) were undefined [R Core Team, 2024]. Unlike Stata’s system, which was handling such conditions seamlessly, running it in R required explicit if/else logic. The analysis also produced expected warnings about mass points in the running variables (population counts) which is a common occurrence in administrative data due to rounding and reporting conventions. These warnings were diagnostic rather than problematic, as the *rdrobust* package was able to accommodate such data features.

Table 2: First Stage and Baseline of Baseline Village Characteristics (Replicated)

Index	V1	V2	V3	V4	V5	V6	V7
	First stage GP pop.	Baseline covariates 1991 Primary school	Middle school	Paved road	Electrified	SC pop.	Literate pop.
Panel A. First stage and balance of characteristics (1991) RD Estimate	-360.9765*** (27.7138)	-0.0077 (0.0143)	0.0091 (0.0073)	0.0190 (0.0144)	0.0084 (0.0104)	-0.0067 (0.0047)	0.0014 (0.0042)
	[0.0000]	[0.5922]	[0.2109]	[0.1876]	[0.4227]	[0.1480]	[0.7378]
Dep var mean	1777.6956	0.7002	0.0631	0.3986	0.1780	0.2510	0.2741
Bandwidth	113	225	376	287	360	301	183
Effective observations	8,769	18,146	31,287	23,413	29,696	24,743	14,665
Panel B. First stage and balance of characteristics (2011) RD Estimate	-649.0284*** (28.3058)	0.0026 (0.0124)	-0.0026 (0.0127)	0.0076 (0.0135)	0.0222 (0.0146)	0.0027 (0.0056)	0.0038 (0.0034)
	[0.0000]	[0.8320]	[0.8355]	[0.5750]	[0.1291]	[0.6298]	[0.2602]
Dep var mean	2294.1805	0.7515	0.2839	0.6434	0.5719	0.2542	0.5643
Bandwidth	353	252	331	295	280	271	208
Effective observations	28,277	20,361	26,503	23,582	22,264	21,765	16,769

Table 3: Effects on Village Amenities (Replicated)

Index	V1	V2	V3	V4	V5	V6	V7
	Index	2001 Primary school	Middle school	2011 Primary school	Middle school	Above Primary Ed	2019 Education score
Panel A. Educational outcomes RD Estimate	0.0871*** (0.0214)	0.0271** (0.0120)	0.0174* (0.0094)	0.0186 (0.0116)	0.0469*** (0.0147)	13.4027* (7.9380)	1.6716*** (0.4298)
	[0.0000]	[0.0244]	[0.0646]	[0.1073]	[0.0014]	[0.0913]	[1e-04]
Dep var mean	-0.1137	0.8337	0.1203	0.8451	0.4264	570.8651	31.2163
Bandwidth	214	192	391	216	288	370	402
Effective observations	16,894	15,094	31,938	17,146	23,014	29,016	32,564
Panel B. Village-level infrastructure RD Estimate	0.0519*** (0.0195)	0.0306*** (0.0115)	0.0160 (0.0140)	-0.0069 (0.0112)	0.0266** (0.0123)	0.0203 (0.0128)	
	[0.0076]	[0.0078]	[0.2539]	[0.5364]	[0.0298]	[0.1120]	
Dep var mean	0.2631	0.5601	0.3918	0.6865	0.7704	0.3251	
Bandwidth	335	448	316	405	227	339	
Effective observations	27,135	36,995	24,763	33,062	17,949	27,401	
Panel C. Household-level infrastructure RD Estimate	0.0918*** (0.0228)	0.0390** (0.0161)	0.0391 (0.0270)	3.2510*** (0.6413)	-3.3343*** (0.6656)	1.2124*** (0.4631)	
	[1e-04]	[0.0155]	[0.9571]	[0.0000]	[0.0000]	[0.0089]	
Dep var mean	-0.0463	0.0863	20.3090	17.8222	81.0760	6.9791	
Bandwidth	248	299	368	225	225	261	
Effective observations	19,768	24,035	29,919	17,782	17,867	20,848	

Further implementation differences were encountered when re-implementing the *MultiPanelTabRD* function. For example, Stata’s `esttab` command natively supports column group headers (via the `mggroups` option), whereas the R implementation required custom code to replicate this feature. Manual parsing of group labels and pattern vectors was introduced to enable similar formatting functionality for the final output.

Table 4: Effect on Delivery of Services (Replicated)

Index	V1	V2	V3	V4	V5	V6	V7	V8	V9
	Programs Index	BPL Card	Health insur.	Pension	Saubhagya	LPG	Housing benefits	PMMVY	Jan Dhan
Panel A. Individual-level programs									
RD Estimate	0.0868*** (0.0331) [0.0087]	4.9986** (2.4154) [0.0385]	2.2116** (1.0643) [0.0377]	-1.3421 (1.0836) [0.2155]	2.4877* (1.3675) [0.0689]	5.6049** (2.4474) [0.0220]	5.5259** (2.1746) [0.0110]	0.0994 (0.1345) [0.4601]	4.9177** (2.0336) [0.0156]
Dep var mean	0.0993	92.8752	22.3268	35.2728	41.7343	95.8732	41.5092	2.5692	69.2872
Bandwidth	160	318	273	280	394	210	203	364	216
Effective observations	12,892	25,283	21,789	22,411	31,283	16,927	16,321	28,913	17,318
Panel B. Workfare program implementation									
RD Estimate	0.1616*** (0.0258) [0.0000]	0.0114*** (0.0013) [0.0000]	0.3102*** (0.0504) [0.0000]	63.6717*** (9.5927) [0.0000]	22.8877*** (5.4677) [0.0000]	0.0055*** (0.0015) [2e-04]			
Dep var mean	0.0787	0.0780	2.0333	393.2044	147.8036	0.0862			
Bandwidth	337	370	326	320	310	323			
Effective observations	118,649	130,688	115,169	112,979	109,132	113,971			

Collectively, these improvements ensure that the R implementation of the function not only reproduces the statistical results of the Stata version but does so with greater robustness, transparency, and flexibility, thereby supporting more reliable and reproducible research workflows.

Note: Table 4 data is not available and needs to be regenerated.

Table 5: Effects on GP Election Outcomes (Replicated)

Index	V1	V2	V3	V4	V5	V6
	Gram Sabha Total partic.	SC partic.	Female partic.	Election turnout	candidates_pc	effective_candidates_pc
Panel A. Political participation						
RD Estimate	0.0207*** (0.0027) [0.0000]	0.0229*** (0.0046) [0.0000]	0.0028*** (5e-04) [0.0000]	0.1033 (0.6320) [0.8701]	-0.1158 (0.1237) [0.3491]	-0.0663 (0.0776) [0.3931]
Dep var mean	0.0754	0.1070	0.0116	71.0567	4.1184	2.3287
Bandwidth	386	304	391	228	323	292
Effective observations	28,524	21,611	29,048	6,333	9,027	8,160
Panel B. Candidate characteristics						
RD Estimate	0.0908 (0.0704) [0.1967]	0.0080 (0.0161) [0.6212]	0.5774*** (0.2217) [0.0092]	-0.0142 (0.0172) [0.4072]	0.0066 (0.0180) [0.7120]	-0.0265 (0.2407) [0.9124]
Dep var mean	2.4283	0.1124	12.3026	0.2800	0.4470	41.7617
Bandwidth	212	468	157	316	275	351
Effective observations	5,846	6,475	2,175	8,838	7,734	9,781
Panel C. Winner characteristics						
RD Estimate	7e-04 (0.0499) [0.9880]	0.0147 (0.0205) [0.4738]	-0.0600 (0.2253) [0.7901]	-0.0065 (0.0096) [0.4974]	-0.0147 (0.0123) [0.2312]	0.6558** (0.2899) [0.0237]
Dep var mean	2.5064	0.1073	12.6891	0.2517	0.4334	41.1681
Bandwidth	278	356	319	409	293	251
Effective observations	28,729	4,918	4,417	43,265	30,562	25,934

During replication, I initially attempted to use fixed, rule-of-thumb bandwidths (± 200 or ± 300 population units), but this approach produced results inconsistent with the original findings. Upon examining the code more closely, I discovered that the automated MSE-optimal bandwidth selector in `rdrobust` was selecting much narrower windows than my fixed bandwidth attempts, demonstrating the superiority of data-driven bandwidth selection over arbitrary choices. This automated approach frequently chose bandwidths optimized for the specific data characteristics, occasionally reducing effective observations substantially but doing so to minimize mean squared error and maximize the precision of local treatment effect estimation.

The reduction in sample size occurred because the algorithm identified the optimal trade-off between bias and

variance by focusing on observations closest to the threshold where the treatment effect is most precisely estimated. Rather than being a limitation, this reflects the automated selector’s ability to identify the most informative data points for causal inference. The fact that estimates in Tables 1, 2, and 3 remained robust across both bandwidth approaches, rather than becoming unstable, demonstrates that the automated method successfully isolated genuine treatment effects while eliminating noise from observations too distant from the cutoff. This consistency validates the automated bandwidth selection’s ability to capture stable, local treatment effects with greater precision than fixed bandwidth approaches. The automated method’s focus on MSE optimization ensures that the observed governance improvements from smaller polity size reflect the most statistically efficient estimates of genuine policy effects, free from the arbitrary assumptions inherent in rule-of-thumb bandwidth selection.

To investigate discrepancies in effective observations across Regression Discontinuity (RD) specifications for Table 6, a structured diagnostic was conducted using the variable `ma_indexXpv10_0` as a representative test case. The first step involved checking the availability of complete cases for the dependent variable, the running variable (`pc11_pca_tot_p`), and the clustering variable (`gp_code_lgd21_num`). This revealed that while the total dataset contained a large number of observations, a considerable share was dropped due to missing values in one or more of these required fields, resulting in fewer complete cases eligible for RD analysis.

4. Conclusion

Overall, the main findings proved highly replicable across different software implementations and methodological approaches. Despite initial challenges with bandwidth selection and software-specific functions, the core regression discontinuity estimates remained consistent, demonstrating the robustness of the underlying empirical relationships. This provides strong validation of the original study’s conclusions about the causal effects of polity size on local governance performance.

However, external validity and generalisability remain significant concerns. The analysis centers specifically on Uttar Pradesh, a state characterized by low bureaucratic capacity, high poverty, and substantial variation in local governance quality. Given these distinctive institutional and socioeconomic contexts, the estimated effects of polity size might not generalize easily to other Indian states with higher bureaucratic capacity, more effective governance, or differing cultural and political contexts. Moreover, Uttar Pradesh’s unique rule for GP delimitation distinguishes it from other states’ approaches. This raises questions about whether the observed benefits of smaller polity size reflect this institutional design rather than a universal relationship between local government size and governance quality across India’s diverse panchayati raj systems.

Moreover, the authors successfully identify the Local Average Treatment Effect (LATE) at the specific threshold of 1,000 residents, providing precise causal estimates. Their supplementary analyses also explores some variation around this threshold, which helps establish a strong foundation for understanding how polity size affects governance outcomes. Their findings demonstrate that the current institutional design works effectively, though future researchers could build on these findings by examining whether similar benefits extend across different population ranges. Future studies could explore heterogeneity at multiple thresholds or examine continuous variation in GP sizes to offer more comprehensive guidance on optimizing polity size for governance quality across diverse contexts.

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